Foundations of Data Analysis

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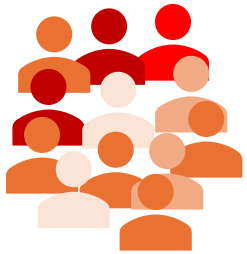


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What is data analysis?

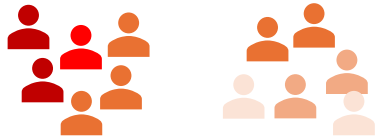
You have collected data from 100 patients.
What do you do next?



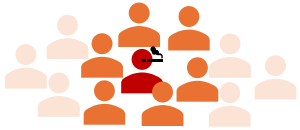
Research question → Data → Analysis → Results → Interpretation



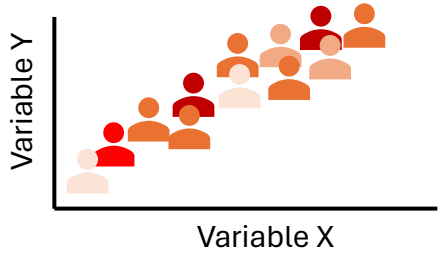
Describe what we observe



Compare groups



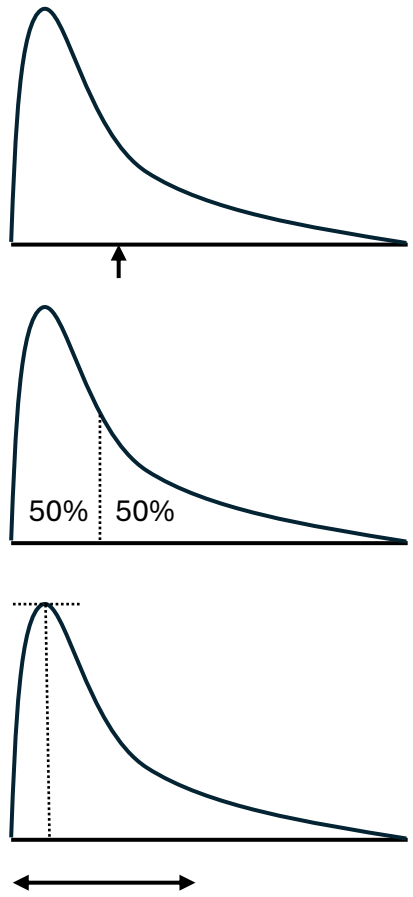
Identify themes and meanings



Look for associations / patterns

Describing data: Summary Statistics

Statistic	What does it tell us?
Mean ($\bar{x}$)	It is the average, found by adding all values and dividing by the number of values. It gives an overall measure but can be affected by extreme values.
Median	It represents the middle value of a data set when the numbers are arranged in order, and it isn't easily influenced by unusually high or low numbers.
Mode	It is the most frequent value in the dataset. These measures help summarize data and provide insights into its central characteristics.
Range	Spread from minimum to maximum



Example data:
1 1 2 3 4 5 6 7 8 9 10

- Mean:
- Median:
- Mode:

Which one to chose?

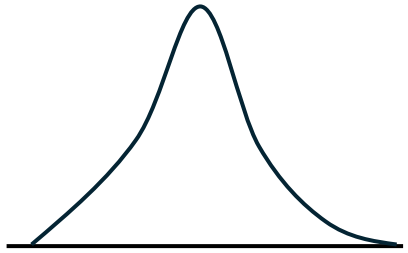
Sallaries in a country
Incidence of desease
Most common dosis prescribed.

What summary statistics shoud you choose?

Continuous & Symmetrical

Normally distributed health data (e.g. blood pressure, patient height).

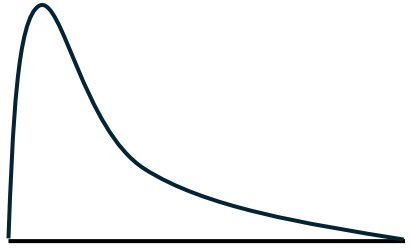
Optimal Choice: Mean



Highly Skewed Data

Data with extreme values (e.g. healthcare costs, disease incidence, national salaries).

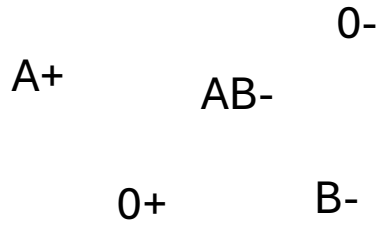
Optimal Choice: Median



Categorical & Nominal

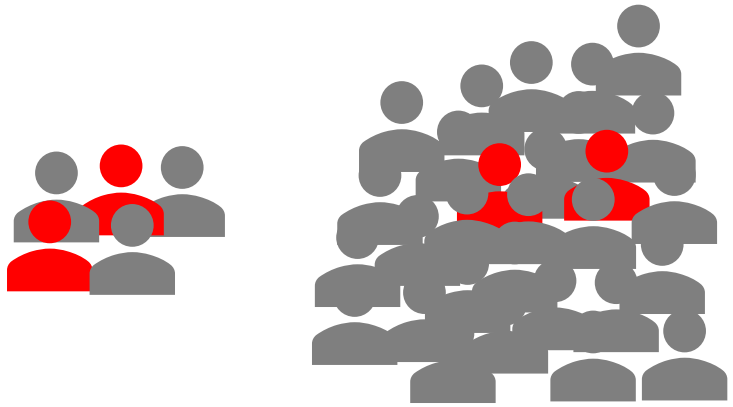
Discrete categories (e.g. blood groups, most common medication dose prescribed).

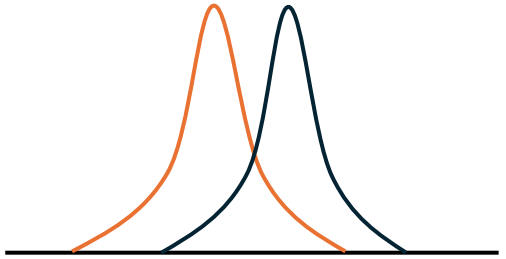
Optimal Choice: Mode



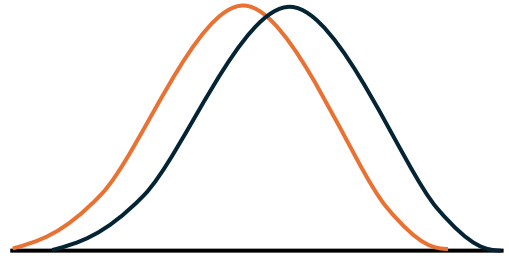
Describing data: Sample size and spread

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$



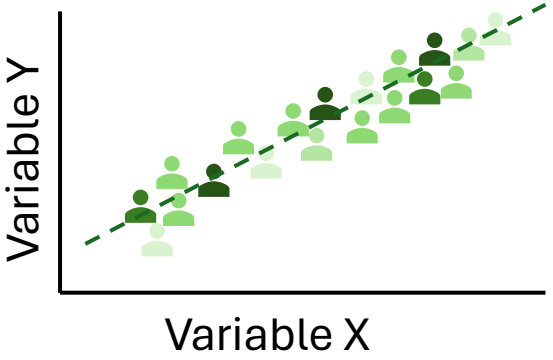


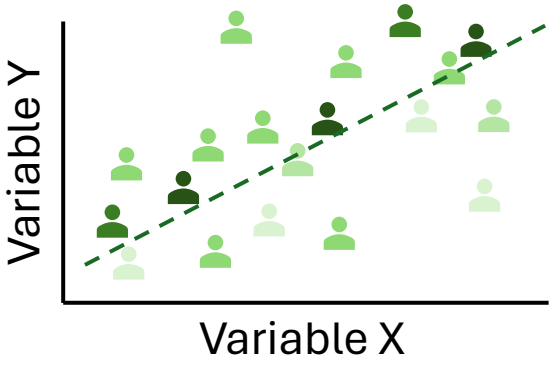
Low variability = Relatively similar measurements



High variability = More spread out data

Sample size (n) is determined by how big an effect you want to detect, how much variability there is in the data, and how certain you want to be that you will detect that effect.

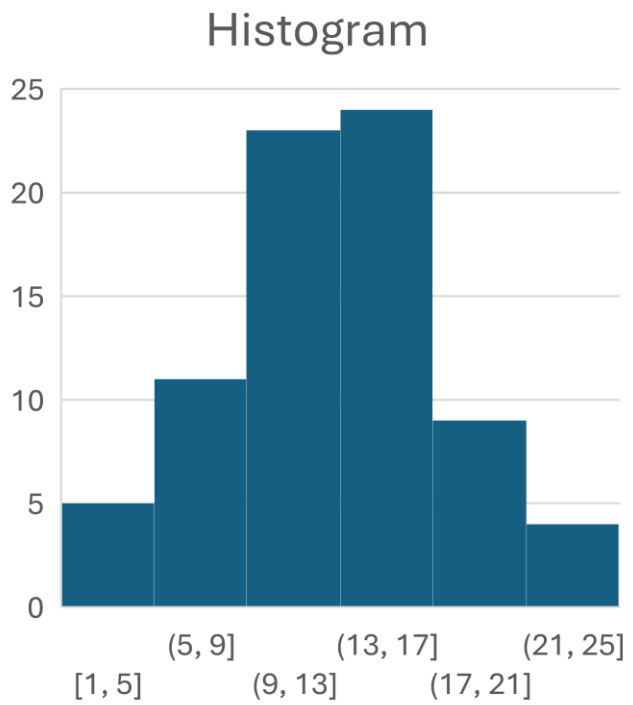




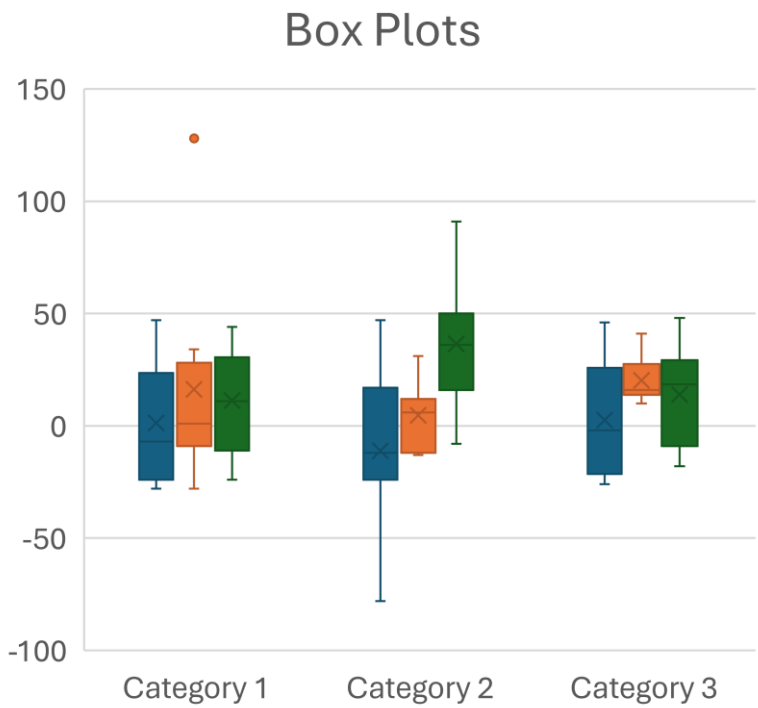
Presentation of Data

Variable	N	Mean	SD	P25	P50	P75	Min	Max
CSR	36,436	2.59	3.510	0	2	3	0	33
community CSR	36,436	0.210	0.570	0	0	0	0	6
governance CSR	36,436	0.390	0.630	0	0	1	0	6
diversity CSR	36,436	0.750	1.060	0	0	1	0	8
employee CSR	36,436	0.550	0.990	0	0	1	0	10
environmental CSR	36,436	0.360	0.960	0	0	0	0	9
human rights CSR	36,436	0.0600	0.270	0	0	0	0	4
product CSR	36,436	0.260	0.630	0	0	0	0	6
bonus-diversity	30,132	0.330	0.210	0.210	0.260	0.340	0.0900	1
salary-diversity	36,357	0.220	0.0700	0.180	0.210	0.230	0.0700	1
stock-diversity	16,847	0.410	0.230	0.260	0.320	0.460	0.130	1
tenure-diversity	35,478	0.790	0.240	0.530	1	1	0.130	1
age-diversity	26,182	0.430	0.300	0.200	0.330	0.510	0.0900	1
gender-diversity	36,436	0.960	0.190	1	1	1	0	1
firm size	36,436	6.060	28.51	0.310	0.960	3.520	0	2491
PPE	34,883	0.280	0.240	0.0800	0.210	0.430	0	0.980
R&D	36,436	0.100	3.410	0	0	0.0300	0	434.0
profitability	36,267	0.100	0.120	0.0500	0.0900	0.150	−2.690	0.870
leverage	36,347	0.540	0.260	0.360	0.550	0.700	0.0100	4.070
cash	36,425	0.140	0.180	0.0200	0.0700	0.200	0	0.970

Presenting the raw numbers is necessary for transparency, but it is a nightmare for readability.

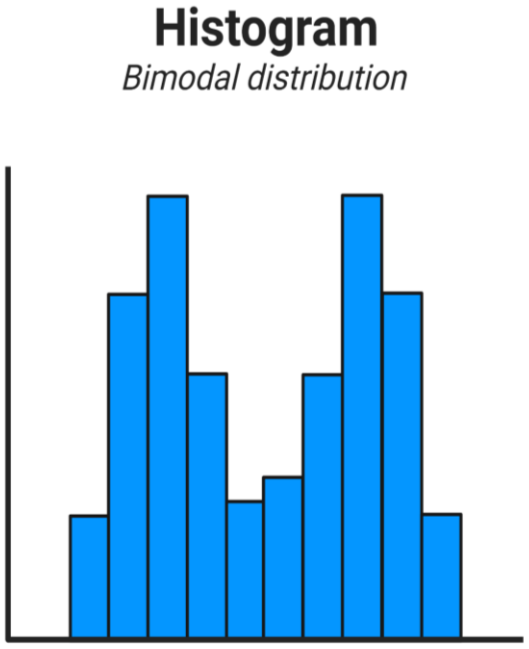


Displays continuous variable distributions. Identifies modal peaks, symmetry, and skewness (right vs. left tail elongation).



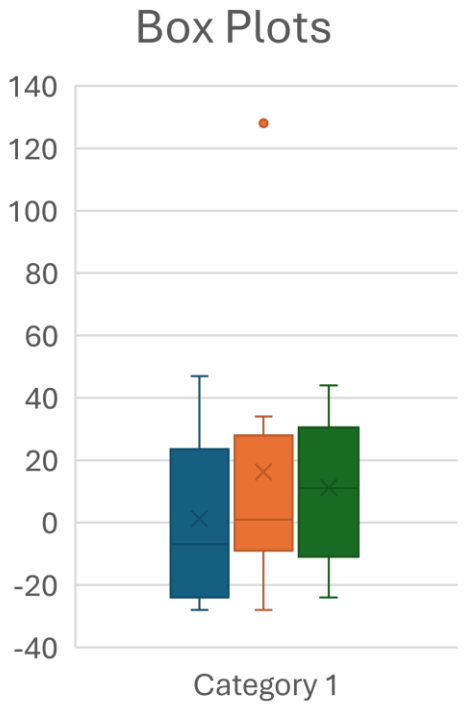
Highlights 5-number summary (Min, Q1, Median, Q3, Max). Instantly reveals Interquartile Range (IQR) and extreme clinical outliers.

Unusual-looking data



If the data is bimodally distributed
This means there are **two distinct peaks**, which may indicate that your data contains **two different groups**.

- **Check why:** Look for a meaningful reason for the two groups (e.g., age groups, two treatments, different populations).
- **Don't automatically combine them:** Combining the groups can hide important differences.
- **Consider analyzing the groups separately** if there is a logical reason to do so.



If you observe an extreme outlier

- **Check for an error:** It could be a measurement, data-entry, or recording mistake.
- If it's an error → **correct or remove it** if justified.
- If it's a real observation → **don't automatically remove it**.
- Check how much it affects your results. You can consider reporting the analysis **with and without the outlier**.
- If the data are strongly affected, **robust statistics** (e.g., median) may be useful.

Don't remove data simply because it looks unusual. First understand **why** it is unusual.

Comparing groups

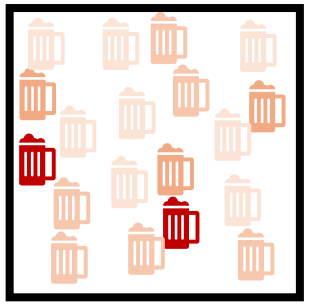


Group	Beer productivity
Treatment A	302.7 L
Treatment B	509.6 L

Group	Mean pain score
Treatment A	4.2
Treatment B	6.1





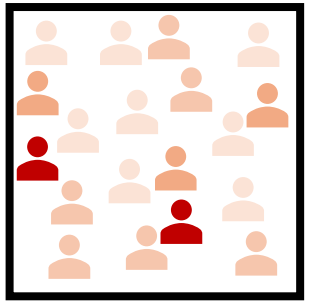


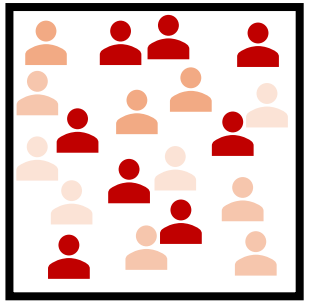


The **t-test** allows the comparison of **means** between two groups.

Does adding more barely into the process result in more beer production?

Do patients receiving Treatment A have lower pain scores than patients receiving Treatment B?





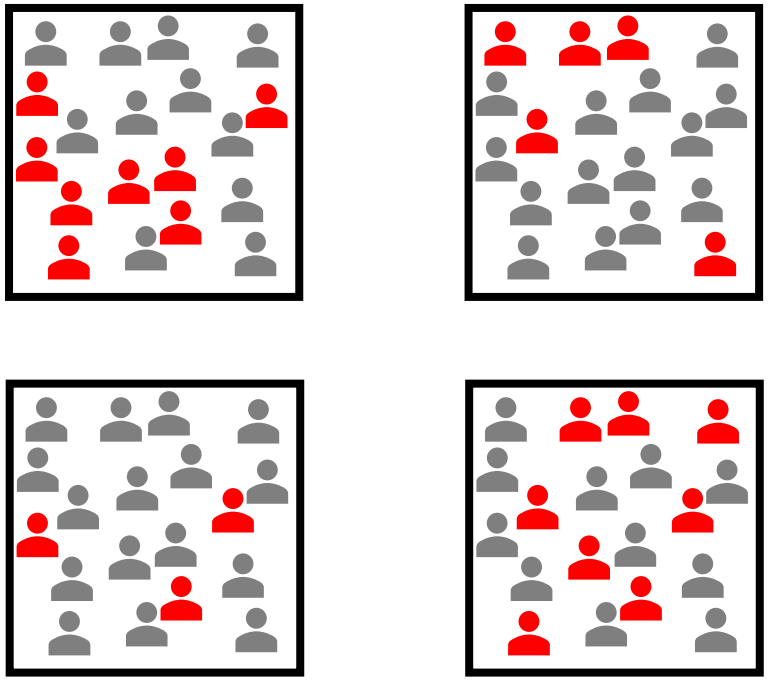
$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

t = signal ÷ noise

Independent variable: **Numeric**
Independent variable: **Categorical**

Chi-square

	Hypertension	No hypertension
Smokers	30	20
Non-smokers	15	35



Is smoking status associated with having hypertension?

Chi-square is useful for testing whether there's an association between **categorical** variables by comparing observed counts to expected **counts.**

The key distinction:

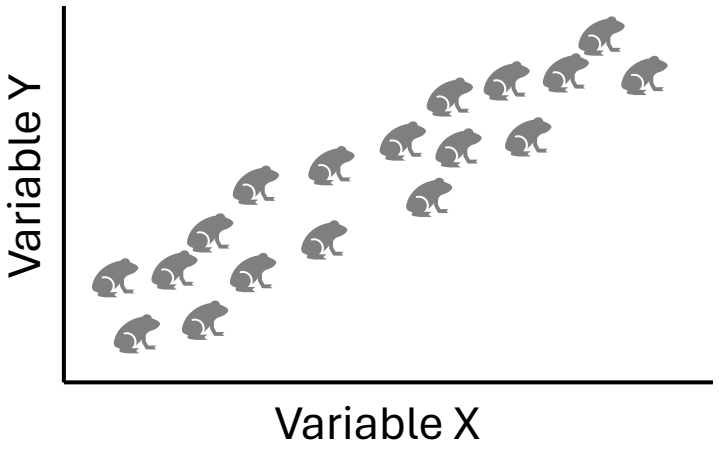
- **t-test:** compares means
- **Chi-square:** compares categorical counts/proportions

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Independent variable: **Numeric**
Independent variable: **Categorical**

Correlation

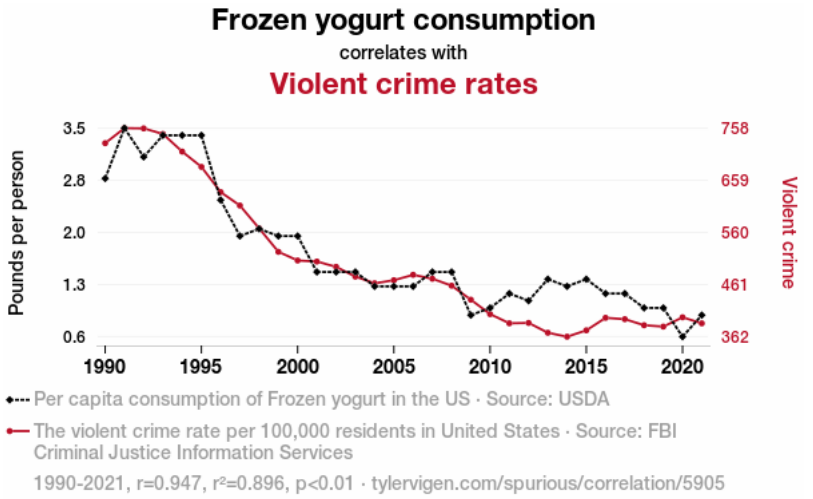
Poison dart frog	Beetle intake (beetles per day)	Skin alkaloid concentration (µg/g)
A	2	18
B	4	31
C	6	47
D	8	59
E	10	72



Correlation examines the relationship between **two numeric variables**.

Independent variable: **Numeric**
Independent variable: **Numeric**

⚠ Correlation doesn't mean causation





How do we interpret a p-value?



The null hypothesis (H_0) usually represents no effect, no difference, or no association.

The hypothesis (H_1) is a testable statement predicting that there is a relationship or difference between variables.

The **p-value** helps us assess how compatible our data are with the null hypothesis.

“The p-value is the probability of obtaining results at least as extreme as those observed, assuming that the null hypothesis is true.”





“The p-value tell us how likely is it to get a result like this if the Null Hypothesis is true.”

How do we interpret a p-value?





What does this mean?
If there were truly no difference between the groups, a result this extreme would be relatively unlikely under the null hypothesis.

- **p = 0.03** → evidence against the null hypothesis
- **p = 0.40** → little evidence against the null hypothesis



"There is no effect"
"We confirm our hypothesis"

"We reject the null hypothesis."
"We fail to reject the null hypothesis."

Effect size and confidence interval?

Statistical significance ≠ clinical importance

Large Sample Size Risk

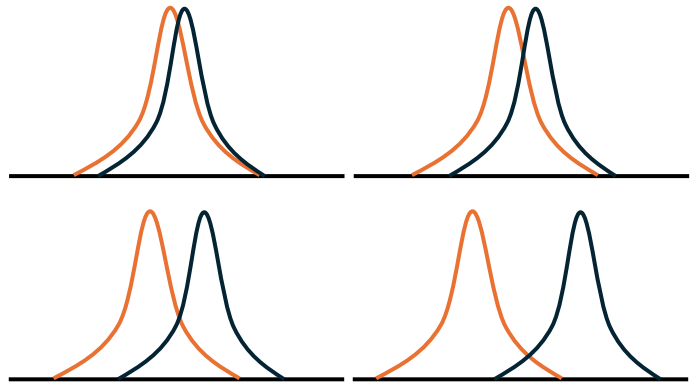
With N = 100,000, a tiny blood pressure drop of 0.2 mmHg achieves **p < 0.001**, but has zero practical clinical value.

Small Sample Size Risk

With N = 15, a major 15 mmHg blood pressure reduction may yield **p = 0.12** due to low statistical power.

Best Practice Solution

Always report **Effect Sizes** (e.g. Cohen's d, Relative Risk) and **95% Confidence Intervals (CI)** alongside p-values to quantify clinical magnitude and precision.



A tiny difference can have a very small p-value with a large sample size.

A clinically important difference may not reach statistical significance with a small sample size.

Qualitative analysis

What if our data are interviews rather than numbers?



→ ***Code: Work Barrier***

What is Coding?

Assigning analytical labels to meaningful text fragments to categorize raw narrative into structured themes.

"I found it difficult to manage medication because I was working full-time." → ***Code: Work Barrier***

Inductive Coding

BOTTOM-UP

Codes and thematic categories emerge directly from raw data without predefined assumptions.

- Forgetting medication during busy shifts
- Feeling embarrassed taking pills publicly
- Changing routines around medication

Data → Emergent Codes

Deductive Coding

TOP-DOWN

Start with predefined concepts, theories, or research frameworks applied to transcript data.

Predefined Codebook Framework:

- Category 1: Medication Access
- Category 2: Work-related Barriers
- Category 3: Work-related barriers
- Category 4: Family support

Theory / Codebook → Applied to Data

Group exercise: Match the question to the analysis

Research question	Appropriate analysis
What is the average BMI?	
Do two groups differ in mean blood pressure?	Chi-square
Is smoking associated with hypertension?	Mean/median
How do patients experience living with chronic pain?	t-test
What themes emerge from interviews about vaccination?	Deductive coding
Does the interview data fit an existing theoretical framework?	Qualitative coding
What is the most common sallary in a population?	Inductive coding
Are smokers more likely to develop lung cancer than non-smokers?	Mode

Group exercise: Match the question to the analysis

Research question

What is the average BMI?

Do two groups differ in mean blood pressure?

Is smoking associated with hypertension?

How do patients experience living with chronic pain?

What themes emerge from interviews about vaccination?

Does the interview data fit an existing theoretical framework?

What is the most common sallary in a population?

Are smokers more likely to develop lung cancer than non-smokers?

Appropriate analysis

Chi-square

Mean/median

t-test

Deductive coding

Qualitative coding

Inductive coding

Mode

